

What If Every Cozystack Change Became a Commit?

COZYSTACK



Reverse GitOps for platform changes

Simon Koudijs, ConfigButler

Who am I?

[Software|Product|Cloud|AI] engineer

- Worked in startups, consultancy, and SaaS
- Left my job last summer to pursue building my own company



What do I do?

- koudijs.dev
 - consultancy, training
- [ConfigButler](#)
 - startup, open source first
 - helps you build high quality configuration
- [Reverse GitOps](#) pioneer
 - manifest, feel free to comment! 😊

Tenant /

root



Deployed applications

Tenant root · tenant-root

NATS 4

**lastone**
NATS● Ready 1h**my-first-again**
NATS● Ready 5d**test123**
NATS● Ready 5d**test1234**
NATS● Ready 1d

Why these resources?

- 🕒 When? ✅
- 👤 Who? ❌
- 🤔 Why? ❌

Git seems far away.

Your options

- Accept it
- Periodic reset
- Limited access
- Audit files ([kube-api](#) to the rescue)

Your options: GitOps

Flip the interface: **Git is the only way in.**

- YAML files / PRs
- Controllers (Argo, Flux)

But: edits can't go through kubectl, dashboard, MCP. The GUI and the API stop being the front door. Git is your **interface**.

PAAS

Click House

Foundation DB

Harbor

Kafka

Maria DB

Mongo DB

NATS

Open BAO

Postgres

Qdrant

Rabbit MQ

Redis

BACKUPS

Plans

Backup Jobs

Backups

Restore Jobs

ADMINISTRATION

Info

Modules

External IPs

Tenant / root

Deployed applications

Tenant root · tenant-1234

+ Deploy application

NATS 4

lastone

NATS

Ready

1h

my-first-again

NATS

Ready

5d

test123

NATS

Ready

5d

test1234

NATS

Ready

1d

8



[https://demo.configbutler.ai/login?
next=/answer/cozysummit-
warmup&code=abc94689412](https://demo.configbutler.ai/login?next=/answer/cozysummit-warmup&code=abc94689412)




```
apiVersion: examples.configbutler.ai/v1alpha1
kind: QuizSubmission
metadata:
  name: cozysummit-warmup-dgm2x
  namespace: voter-production
spec:
  answers:
    - questionId: q1
      singleChoice: Heard of it
    - multiChoice:
      - kubectl
      - Git (PR-based)
      questionId: q2
    - number: 6
      questionId: q3
  sessionRef:
    group: examples.configbutler.ai
    kind: QuizSession
    name: cozysummit-warmup
  submittedAt: "2026-05-26T13:32:56.779Z"
```

Why these resources?

- 🕒 When? ✅
- 👤 Who? ❌
- 🤔 Why? ❌

What if?

What if **every platform change** automatically appeared as clean YAML in Git?

- Fully automated
- Identity
- A silent layer
- API activity → durable, readable history of intent

Reverse GitOps

Classic GitOps: **Git** → **Platform**

Reverse GitOps: **Platform** → **Git**

Git stops being the *interface*.

Git becomes the *memory*.

Reverse GitOps

API-first. Git-recorded. GitOps deploys.

[Read the manifesto ↓](#)

★ Star on GitHub 4

 · I'm speaking at [CozySummit Virtual 2026](#) · [View talks →](#)

Also spoke at: [KubeCon + CloudNativeCon Europe 2026](#) · [Platform Engineering Day](#)

An API-first operating model where users and agents submit validated intent through a typed API, while that intent is automatically recorded in Git as clean declarative YAML manifests for audit, promotion, and distribution.

1. API First
2. Capture Intent, Not Implementation
3. GitOps Still Applies

[reversegitops](#)

 **gitops-reverser** Public

Edit Pins

Watch0

Fork1

Starred11

main

35 Branches

42 Tags

Go to file

Add file

Code

 github-actions[bot] chore(main): release gitops-reverser 0.26.3 (#158) 609 Commits	f74e521 · 9 hours ago
.claude docs: getting started hints for aggregated API usage last week	
.devcontainer fix: ignore subresource audit events (that was hard to de... 3 days ago	
.github fix: ignore subresource audit events (that was hard to de... 3 days ago	
.vscode fix: One edgecase where a secret is deleted (is now recrea... 2 months ago	
api/v1alpha1 fix: showing spec.message at commitrequests 17 hours ago	
charts/gitops-reverser chore(main): release gitops-reverser 0.26.3 (#158) 9 hours ago	
cmd chore: queue defaults (don't grow forever) 18 hours ago	
config fix: showing spec.message at commitrequests 17 hours ago	
docs chore: queue defaults (don't grow forever) 18 hours ago	
hack feat: deduplicated audit ingestion for aggregated API, re... last week	
internal chore: queue defaults (don't grow forever) 18 hours ago	
test fix: allow generated names with a commit request (#155) 19 hours ago	
.dockerignore feat: Use sops to encrypt secrets 3 months ago	
.env.example fix: ignore subresource audit events (that was hard to de... 3 days ago	
.gitignore feat: bumping apiservice-audit-proxy and it's all TLS now last week	

About

Reconciles Kubernetes API resources into a git destination as clean deployable YAML files (audit trail through git commits).

audit-log

gitops

config-as-data

kubernetes-resource-model

reverse-gitops

Readme

Apache-2.0 license

Contributing

Security policy

Activity

Custom properties

11 stars

0 watching

1 fork

Audit log

Report repository

Releases42

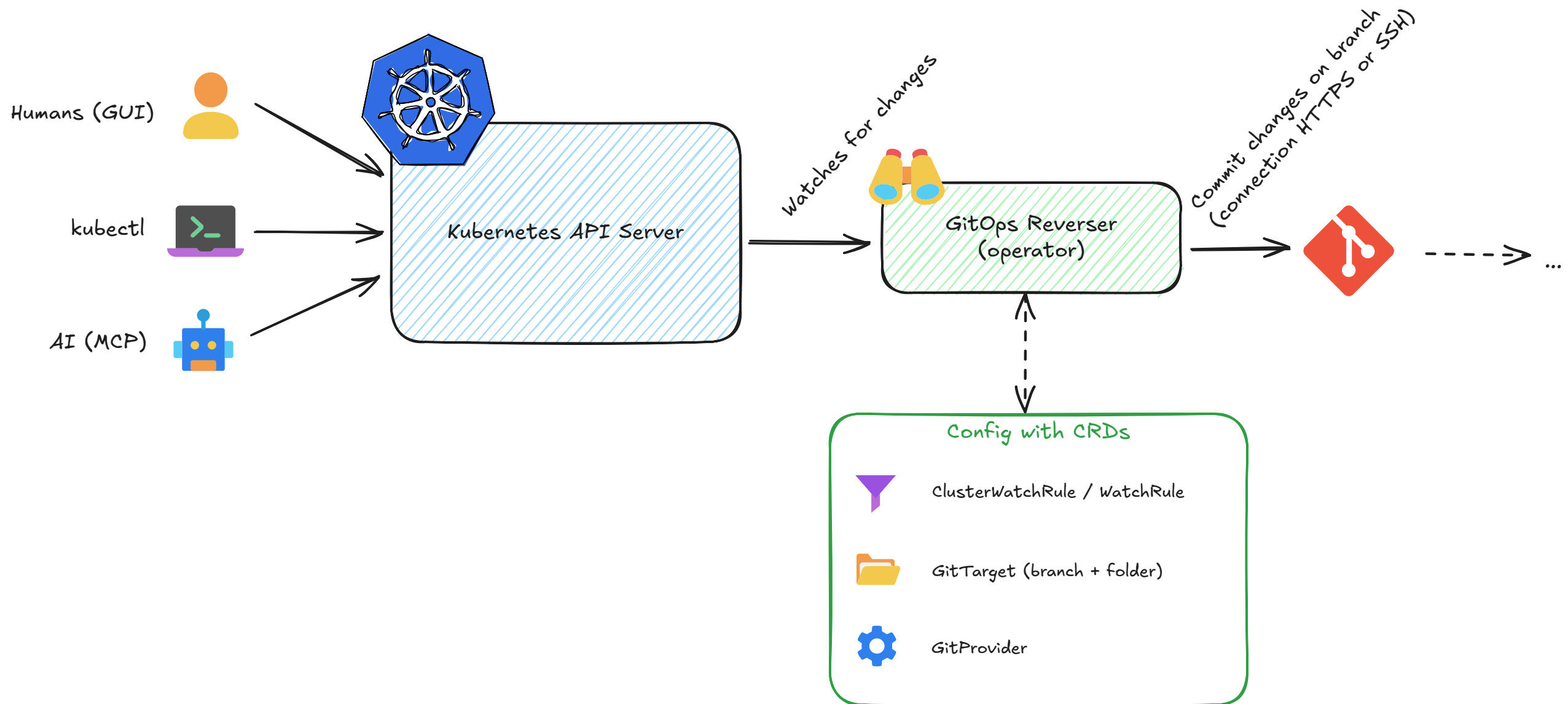
gitops-reverser: v0.26.3

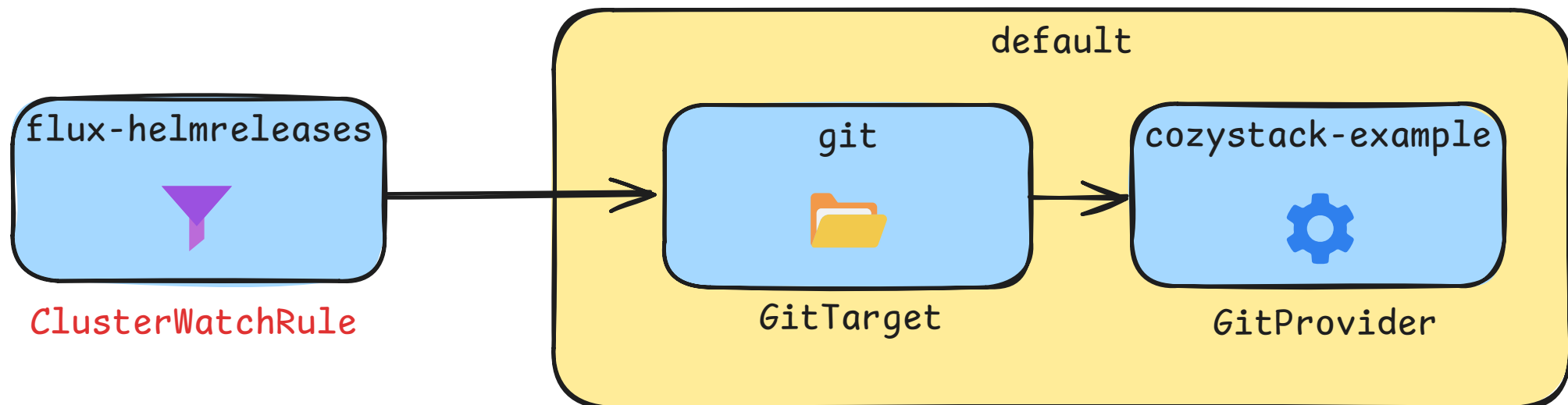
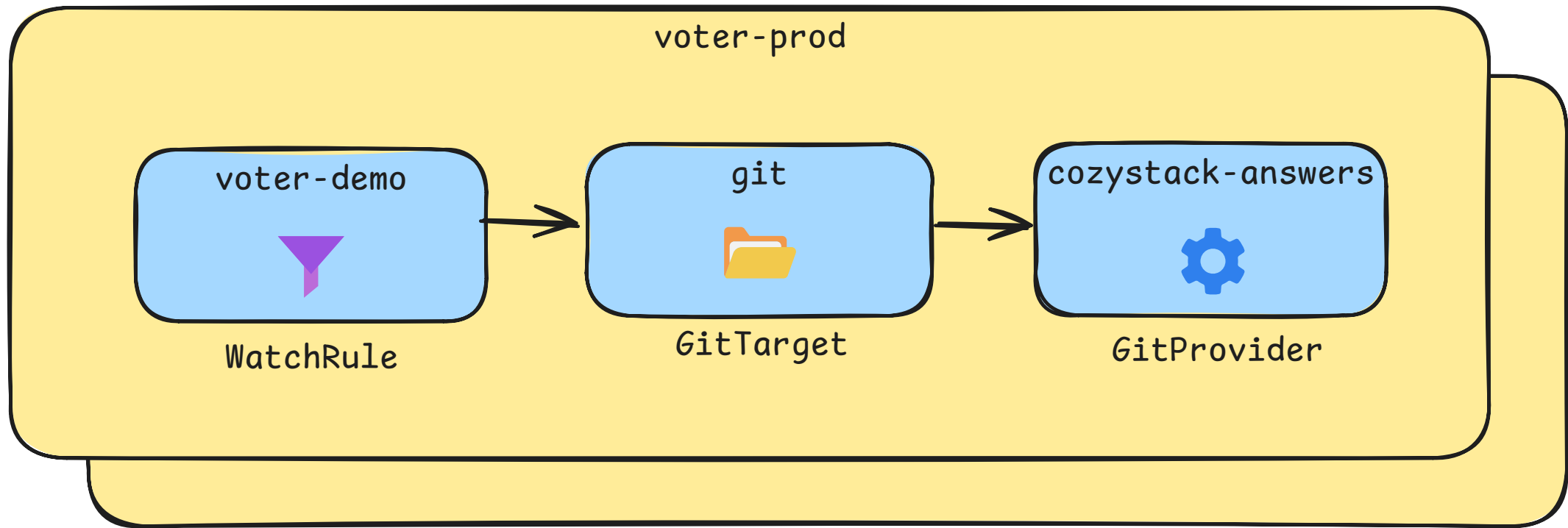
Latest

9 hours ago

+ 41 releases

15





AC

admin@reverser-cozy

apps.foundationdb.org

autoscaling.k8s.io

backups.cozystack.io

cert-manager.io

cilium.io

clickhouse-keeper.altinity.com

clickhouse.altinity.com

configbutler.ai

Cluster Watch Rule

Commit Request

Git Provider

Git Target

Watch Rule

core.strimzi.io

GitTarget

3 items

All namespaces

GitTarget search ...

☐	Name	Namespace	Provider	Branch	Path	Ready	Status	Age
☐	cozystack-example	gitops-reverser	cozystack-example	main	demo	True	All lifecycle gates sati...	6d2h
☐	voter-demo	voter-production	voter-demo	main	voter	True	All lifecycle gates sati...	25h
☐	voter-demo	voter-test	voter-demo	features/some-feature	voter	True	All lifecycle gates sati...	25h

Commits

 main	 All users
🔗 Commits on May 26, 2026	
[CREATE] examples.configbutler.ai/v1alpha1/quizsubmissions/cozysummit-warmup-8dr67  Anonymous 186362 authored and configbutler-bot committed 2 hours ago	Verified
[CREATE] examples.configbutler.ai/v1alpha1/quizsubmissions/cozysummit-feedback-frbf4  Anonymous 146001 authored and configbutler-bot committed 2 hours ago	Verified
[CREATE] examples.configbutler.ai/v1alpha1/quizsubmissions/cozysummit-warmup-dgm2x  Anonymous 146001 authored and configbutler-bot committed 2 hours ago	Verified
reconcile: sync 6 resources  configbutler-bot committed 3 hours ago	Verified
[UPDATE] examples.configbutler.ai/v1alpha1/coffeeconfigs/app-config (#1)  sunib authored 3 hours ago	Verified
[UPDATE] examples.configbutler.ai/v1alpha1/coffeeconfigs/app-config  sunib authored and configbutler-bot committed 3 hours ago	Partially verified
Normal shopname	

Git commits

```
> git log -1 --pretty=fuller
commit 0c242f2fd35d89d2189a6be849220ccce6cc60e2 (HEAD -> main, origin/main, origin/HEAD)
Author:      Anonymous 146001 <146001@demo.configbutler.ai>
AuthorDate:  Tue May 26 13:35:24 2026 +0000
Commit:      ConfigButler bot <286979421+configbutler-bot@users.noreply.github.com>
CommitDate:  Tue May 26 13:35:24 2026 +0000
```

```
[CREATE] examples.configbutler.ai/v1alpha1/quizsubmissions/cozysummit-feedback-frbf4
```

Why these resources?

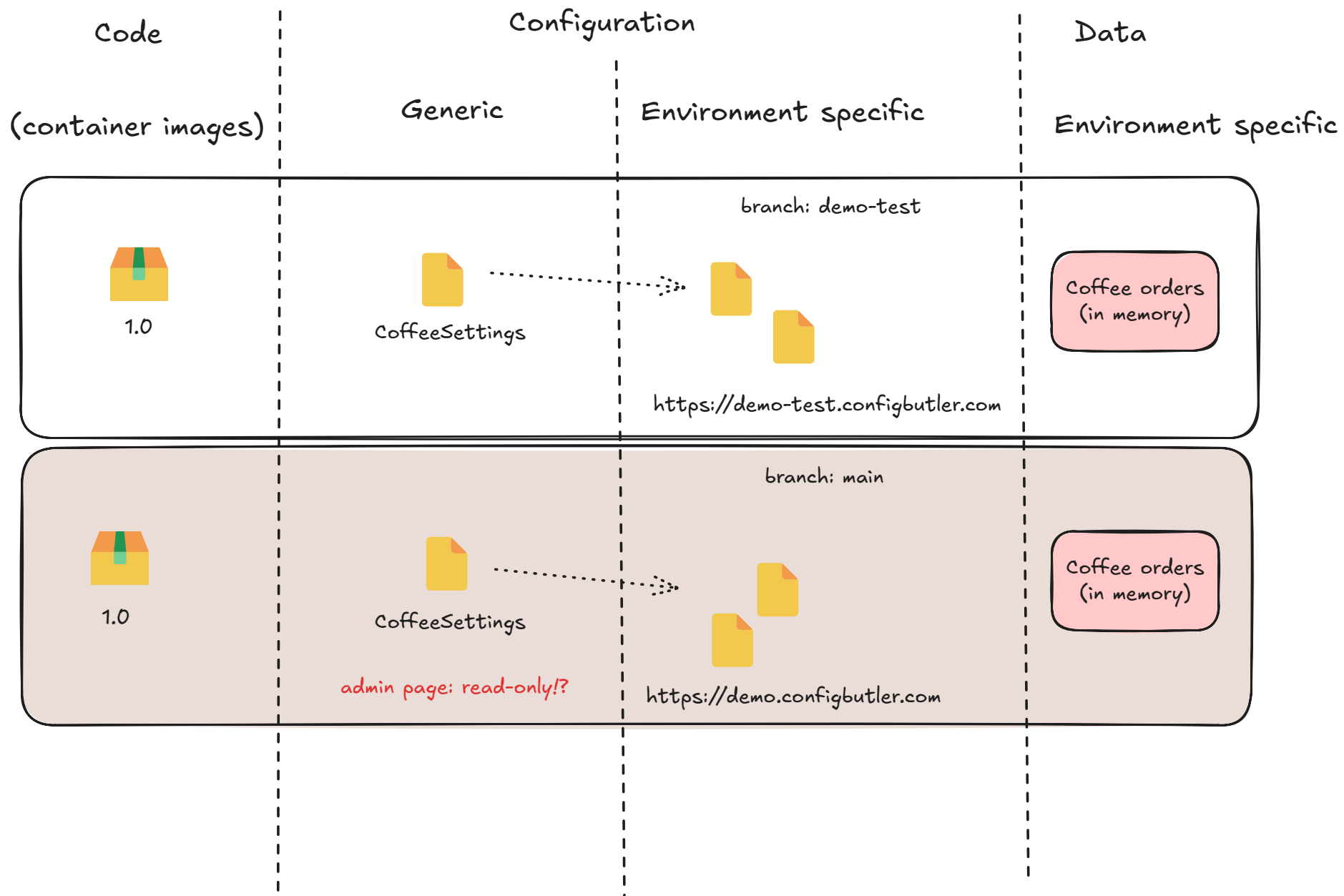
- 🕒 When? ✅
- 👤 Who? ✅
- 🤔 Why? ❌ -> commit message?

Demo part 2

- Let's do some coffee ordering
- Artifacts
- And let's make some changes

demo-test.configbutler.ai

demo.configbutler.ai



COZYSUMMIT COFFEE

Scan the QR code for your free CozySummit coffee

Admin

STANDARD PRICING

CHOOSE YOUR COFFEE

Prices update live. Voucher depletion is only checked when you order.

COFFEE-FLAT-WHITE

FLAT WHITE

Double shot with silky milk

€3.95



COFFEE-ESPRESSO

ESPRESSO

Small, strong, and fast

€2.75



COFFEE-FILTER

FILTER COFFEE

Batch brew for fast queues

€2.50



STOREFRONT CONFIG

GENERATION 13

SHOP NAME

CozySummit Coffee1

CURRENCY

EUR

BANNER TEXT

Scan the QR code for your free CozySummit coffee

PRODUCTS

Add Product

SKU

coffee-flat-white

NAME

Flat White

BASE PRICE (CENTS)

395

☒ Enabled

DESCRIPTION

Double shot with silky milk

€3.95

Remove

SKU

NAME

SIGNED IN

REVIEW AND SAVE

1 CHANGE(S) READY TO SAVE

Yellow dots are local edits. Saving writes your changes into the watched config.

shopName

EDITED

Revert

CozySummit Coffee → CozySummit Coffee1

WHY ARE YOU MAKING THIS CHANGE?

optional for now. Add a short note so people understand what changed and why.

1 edit(s) and 0 missed incoming change(s) in this save.

Save 1 Change

Why these resources?

- 🕒 When? ✓
- 👤 Who? ✓
- 🤔 Why? ✓

Why Cozystack fits

- Kubernetes API first
- GUI uses OIDC
- Talos

Tenant / root

Deployed applications

Tenant root · tenant-root

NATS 4

**lastone**
NATS● Ready 1h**my-first-again**
NATS● Ready 5d**test123**
NATS● Ready 5d**test1234**
NATS● Ready 1d

Commits

main ▾

All users ▾

Commits on May 26, 2026

[UPDATE] helm.toolkit.fluxcd.io/v2/helmreleases/nats-test-system	Verified	a48072
system:serviceaccount:cozy-fluxc... authored and configbutler-bot committed 1 minute ago		
[CREATE] apps.cozystack.io/v1alpha1/natses/test	Partially verified	be762
sunib authored and configbutler-bot committed 1 minute ago		
[CREATE] helm.toolkit.fluxcd.io/v2/helmreleases/nats-test	Verified	03811d
system:serviceaccount:cozy-syste... authored and configbutler-bot committed 1 minute ago		
[DELETE] helm.toolkit.fluxcd.io/v2/helmreleases/nats-lastone-system	Verified	5d26e5
system:serviceaccount:cozy-fluxc... authored and configbutler-bot committed 1 minute ago		
[DELETE] apps.cozystack.io/v1alpha1/natses/lastone	Partially verified	d1af0c
sunib authored and configbutler-bot committed 2 minutes ago		
[DELETE] helm.toolkit.fluxcd.io/v2/helmreleases/nats-lastone	Verified	91cf3a
system:serviceaccount:cozy-syste... authored and configbutler-bot committed 2 minutes ago		
reconcile: sync 108 resources	Verified	293b8d0a
configbutler-bot committed 3 hours ago		

How can this work?

Option	Phase	Identity	Why not enough
Watch	Post	No	Sees resulting state, not who changed it.
Admission webhook	Pre	Yes	Sees the request, not the final persisted outcome.
Audit webhooks	Post-ish	Yes	Best fit: who did what, to which resource, and how the API server responded.

What you gain

- **Auditability** — versioned, diffable, shareable
- **Easier on-ramp to GitOps** — no YAML fight, the API answers fast
- **GUI without guilt** — click in the dashboard, still get a commit
- **Non-engineers can propose changes** — through the tools they already use
- **Test-cluster → PR → prod** — promote what worked

The honest part

- Setup is not easy
 - api-audit-proxy
 - kube-api tuning
- Secrets
 - gitops-reverser can encrypt secrets with SOPS
 - Please keep them in dedicated resources, not inline fields

What's next?

- Please let me know what you think!
- I can't cover everything
- Promotion and review of changes

Takeaways

1. The Kubernetes API is already a record of intent: capture it
2. Reverse GitOps gives auditability without changing how operators work
3. Cozystack's high-level resources make the diffs *human-readable*

Contact details and presentation at <https://koudijs.dev>



<https://demo.configbutler.ai/login?next=/answer/cozysummit-feedback&code=abc94689412>